5) A RESIDENTIAL BUILDING – Using AUTO CAD

The following line sketch shows the internal dimensions of A RESIDENTIAL BUILDING. Draw to a scale of 1:50 / 1:00 the following views: using AUTO CAD

A) Plan

B) Section on AB

C) Elevation

SPECIFICATIONS	
a) FOUNDATION	Depth of foundation is 1.2 m below natural ground level. The concrete base course is 1.2 m wide and 0.3 m thick in PCC of 1:4:8 mix
b) FOOTING	Isolated footings as per design in RCC 1:1.5:3 will be provided in appropriate places. The breadth and thickness of footing will be 0.90 m and 0.45 m respectively. Over the footing, RCC column 0.23 x 0.23m size are raised up to the grade beam level.
c) GRADE BEAM	A beam of size 0.23 x 0.30 m is provided as per the structural design alround the building up to the ground level.
d) BASEMENT	The basement will be of brick masonry in CM 1:5 and of height 0.6m above the Grade beam. The thickness of plinth wall is 0.23m and a damp proof course 0.02m thick in CM 1:3 which is mixed with 5% crude oil will be provide around the building.
e) FLOORING	Over 0.450m depth of sand filling, flooring of 0.15m thick in CC 1:4:8 mix finished with marble stone is provided.
f) SUPERSTRUCTURE	RCC columns 0.23 x 0.23m are raised above the Grade beam upto roof level in correct sequence. The thickness of walls above plinth level is 0.23 in brick work using CM 1:5. The height of the parapet wall is 0.9 m above the roof top level. Lintel – cum – sunshade will be provided over the door and window openings. Thickness and bearing of RCC lintel in CC 1:1.5:3 will be 0.15m. The projection of sunshade will be 0.45m from the wall. The thickness is 0.08m at support and 0.05m at free end. Ceiling height will be 3m above the floor level.
g) ROOFING	Roofing will be of flat RCC in CC 1:1.5:3, 0.12m thick. A weathering course of 0.1m thick is provided over the roof slab with sufficient slope to drain rainwater.
h) STEPS	Tread = 0.3m. Rise = 0.15m.

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